Acta Psychologica 261 (2025) 105791

Contents lists available at ScienceDirect

Acta Psychologica

journal homepage: www.elsevier.com/locate/actpsy

Association between Generative AI self-efficacy and Generative AI acceptance: The mediating role of Generative AI trust and the moderating role of Generative AI risk perception

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ARTICLE INFO

Keywords:
Generative AI self-efficacy
Generative AI trust
Generative AI risk perception
Generative AI acceptance
University students

ABSTRACT

Generative AI has garnered global attention. This study aims to explore the association between generative AI self-efficacy and generative AI acceptance among university students and underlying mechanisms. A sample of 353 university students in China was measured using scales for generative AI self-efficacy, generative AI trust, generative AI risk perception, and generative AI acceptance. Structural equation model and Model 1 from SPSS PROCESS 4.0 were employed to test the research hypotheses. The results indicated that a positive association between generative AI self-efficacy and generative AI acceptance among university students ($\beta = 0.49$, $P < .001$). Assistance, comfort with generative AI, and technological skills are positively association with generative AI acceptance. Generative AI trust mediated the relationship between generative AI self-efficacy and generative AI acceptance, as well as the relationship between anthropomorphic interaction and generative AI acceptance. Generative AI trust also mediated the relationship between technological skills and generative AI acceptance. Moreover, Generative AI risk perception negatively moderated the relationship between generative AI trust and generative AI acceptance. This study provides a theoretical foundation for further research on generative AI.

1. Introduction

Generative artificial intelligence technology is reshaping the paradigm of human-computer interaction through its development and application (Baek et al., 2021; Baek & Kim, 2023). Generative AI provides a way of conversational human-computer interaction, altering traditional models of information retrieval and knowledge production (Baek & Kim, 2023). This technology, relying on vast training data and deep learning algorithms, meets users' needs for conversational interaction and knowledge acquisition (Baek & Kim, 2023; Shankland, 2023). Since the launch of ChatGPT by OpenAI in 2022, there has been a global surge in the research and development of generative AI, with China introducing several localized generative AI products, such as DeepSeek, Tongyi, Doubao, and Kimi. These products can be used for text generation, audio and video content creation, and code writing (Baek & Kim, 2023; China Internet Network Information Center, 2024; Feuerriegel et al., 2024).

University students, as the digital natives, have widely adopted generative AI for information gathering, essay writing, document summarization, and emotional interaction (Gasaymeh et al., 2024; Golding et al., 2024; Nikolopoulou, 2024; Prensky, 2001). This may be because after initial use of generative AI, students find it convenient to use and beneficial to their studies, thereby increasing their trust and acceptance of generative AI (Park & Kim, 2023; Ventre & Kolbe, 2020; Wang & Yu, 2024). Moreover, Generative AI self-efficacy may be also correlated to Generative AI acceptance among university students. Because computer self-efficacy is a widely studied task-specific self-efficacy that affects an individual's technology adoption, and this implies the generative AI self-efficacy may correlate with students' acceptance of generative AI (Igbaria & Iivari, 1995; Wang & Chuang, 2024).

Based on The Technology Acceptance Model, previous studies have confirmed that the positive correlation between self-efficacy, confidence, trust and acceptance (Reid & Levy, 2008; Xu et al., 2024). Moreover, risk perception serves as a risk factor in technology acceptance, where perceived risks can diminish trust and acceptance (Im et al., 2008; Lorenzo Romero et al., 2011). However, it remains unclear whether these findings can also be validated in the context of generative artificial intelligence. In particular, researchers have recently

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https://doi.org/10.1016/j.actpsy.2025.105791
Received 12 February 2025; Received in revised form 13 October 2025; Accepted 13 October 2025
Available online 15 October 2025

introduced the new concept of AI self-efficacy, which is divided into four dimensions: assistance, anthropomorphic interaction, comfort with generative AI, and technological skills (Wang & Chuang, 2024). The relationships between these four dimensions and trust, risk perception, and technology acceptance are still a gap.

Thus, drawing from the Social Cognitive Theory, Technology Acceptance Model and previous studies, this study aims to investigate the relationship between generative AI self-efficacy (assistance, anthropomorphic interaction, comfort with generative AI, and technological skills) and generative AI acceptance among university students. Furthermore, the mediating role of generative AI trust and the moderating effect of generative AI risk perception will be investigated.

2. Literature review and hypothesis development

2.1. Self-efficacy and technology acceptance

Social Cognitive Theory (SCT) was proposed by Bandura, suggesting that self-efficacy is an individual's subjective judgment of their ability to perform behaviors required in specific situations, a concept that plays a crucial role in shaping personal beliefs and behaviors (Bandura, 1982; Compeau & Higgins, 1995; Voskobojnikov et al., 2021). However, given that self-efficacy comprises individuals' perceived beliefs regarding specific activities, its measurement should be tied up with context-specific tasks and scenarios (Latikka et al., 2019). In the technology context, self-efficacy has been examined for the use of the internet, computers, and robots and demonstrated internet self-efficacy, computer self-efficacy, and Robot use self-efficacy were associated with the acceptance to use related technology (Hsia et al., 2014; Hsu & Chiu, 2004; Latikka et al., 2019).

With the rapid development of artificial intelligence technologies, generative AI self-efficacy has gradually emerged as a new research topic of academic interest. Generative AI self-efficacy refers to individuals' overall confidence in their ability to use or interact with Generative AI (Baek & Kim, 2023; Bandura, 1982; Wang & Chuang, 2024). Generative AI self-efficacy is divided into four dimensions: Assistance, anthropomorphic interaction, comfort with AI, and technological skills. Assistance refers to the perceived effectiveness or value of artificial intelligence technologies or products in aiding individuals to accomplish tasks (Davis, 1989; Wang & Chuang, 2024). Anthropomorphic interaction denotes the perceiving human-like traits in AI interfaces (Li & Suh, 2022; Wang & Chuang, 2024). Comfort with AI reflects a person's emotional response and sense of ease when engaging with AI technologies or products (Wang & Chuang, 2024). Technological skills encompass an individual's knowledge base and confidence level in utilizing AI technologies or products (Wang & Chuang, 2024).

According to the Social Cognitive Theory (SCT), generative AI self-efficacy may affect individuals' beliefs and behaviors for generative AI (Bandura, 1982; Compeau & Higgins, 1995). Previous studies have indicated that self-efficacy in AI learning moderates the relationship between job insecurity induced by AI and psychological safety (Kim & Kim, 2024). Specifically, a high level of self-efficacy can effectively mitigate the negative impact of job insecurity on psychological safety (Kim & Kim, 2024). For example, the higher technology self-efficacy will reduce the users' fear of AI; users are more likely to accept AI technologies (Montag et al., 2023). Moreover, the user has good information and communication technology self-efficacy; they received more perceived ease of use and more willingness to use AI chatbots (Esiyok et al., 2024). An empirical study also confirmed the positive correlation between AI self-efficacy and willingness to use AI (Hong, 2022). Thus, we propose:

H1. Generative AI self-efficacy is positively association with generative AI acceptance.

H1a. Assistance is positively association with generative AI acceptance.

H1b. Anthropomorphic interaction is positively association with generative AI acceptance.

H1c. Comfort with generative AI is positively association with generative AI acceptance.

H1d. Technological skills is positively association with generative AI acceptance.

2.2. Mediating effect of AI trust

The Technology Acceptance Model (TAM) was proposed by Davis in 1989, suggesting that the attitude toward use is primarily influenced by two core variables: Perceived ease of use and perceived usefulness (Davis, 1989). The attitude toward using technology affects the behavioral intention, and the behavioral intention further impacts usage behavior (Zainab et al., 2017). Previous studies have used the Technology Acceptance Model (TAM) to explain the technological acceptance, and this model demonstrates strong explanatory power in understanding technology acceptance (Park et al., 2014; Purnomo & Lee, 2013). Some studies also combined the technology self-efficacy and trust into the TAM, suggesting the technology self-efficacy influence on the attitude toward technology through the trust (Reid & Levy, 2008; Xu et al., 2024). This implies that AI trust may be as a mediator between generative AI self-efficacy and generative AI acceptance.

AI trust refers to individuals' positive psychological disposition and confidence toward AI during interactions, rooted in their belief that AI possesses the capability to assist them in attaining defined objectives (Zhang, Hu, et al., 2025). Previous studies show that self-efficacy can enhance trust (Diotaiuti et al., 2021; Na, 2022). Especially when users are confident in their ability to operate and fully understand AI systems, they are more likely to recognize the reliability of the system's output (Baek & Kim, 2023; Morgan & Hunt, 1994). This cognitive belief significantly enhances users' trust in AI (Baek & Kim, 2023; Morgan & Hunt, 1994). Specifically, self-efficacy positively influences users' perceived ease of use of AI systems, which further positively predicts perceived usefulness (Davis, 1989; Song et al., 2022; Yao & Wang, 2024). When users gain positive experiences with AI through continuous use, their perception of the usefulness of AI technology will significantly increase, and this process simultaneously reinforces users' trust in AI technology (Baek & Kim, 2023; Kanont et al., 2024; Song et al., 2022; Venkatesh & Davis, 2000).

Trust is also a predictor of users' attitudes and willingness to use AI (Ali et al., 2024; Choung et al., 2023). For example, in AI-enhanced design use, users have higher trust in AI-enhanced design, appearing to have more acceptance for AI-enhanced design (Zhou, Liu, et al., 2024). Which means that perceived AI trust affects consumer behavioral intentions and evaluations (Kim et al., 2021). Similarly, In the study of automated technology, A positive correlation was confirmed between the propensity to trust in (automated) technology and the acceptance of artificial intelligence (Montag et al., 2023). Therefore, we propose:

H2. Generative AI trust mediates the relationship between generative AI self-efficacy and generative AI acceptance.

H2a. Generative AI trust mediates the relationship between assistance and generative AI acceptance.

H2b. Generative AI trust mediates the relationship between Anthropomorphic interaction and generative AI acceptance.

H2c. Generative AI trust mediates the relationship between comfort with generative AI and generative AI acceptance.

H2d. Generative AI trust mediates the relationship between technological skills and generative AI acceptance.

2.3. Moderating effect of AI risk perception

As industrial society transitions to a post-industrial or "risk society," humanity faces new types of risks induced by technological advancements, globalization, and environmental issues (Beck, 1992). The development and application of AI technology bring risks characterized by a high degree of uncertainty and unpredictability (Beck, 1992; Song et al., 2022; Yampolskiy, 2024). AI risk perception describes an individual's anticipation of possible adverse outcomes resulting from AI implementation (Li, 2024). Global surveys indicate that the public generally views artificial intelligence as a double-edged sword: while AI presents numerous opportunities, its potential risks cannot be overlooked (Floridi et al., 2018; Schwesig et al., 2023). Individuals' perception of the potential opportunities associated with AI can enhance their willingness to use the technology; however, the risks that coexist with these opportunities significantly diminish public trust and willingness to engage with AI (Schwesig et al., 2023; Shao et al., 2024; Song et al., 2022; Zhu et al., 2024). When high risks are involved, users exhibit low trust and negative attitudes toward AI-powered products and services (Cicek et al., 2024). For example, research on pre-service teachers' trust in artificial intelligence has revealed that perceived privacy and security risks diminish trust in AI among pre-service teachers in rural areas (Zhang, Wang, et al., 2025). Gerlich (2023) discovered that individuals who recognize the potential dangers and risks associated with AI often hold a pessimistic view regarding its widespread acceptance and use. This phenomenon is particularly evident among college students, who perceive the risks associated with using AI for learning, such as academic misconduct and bias (Ivanov et al., 2024; Zhou, Liu, et al., 2024). These risks tend to diminish their willingness to utilize AI (Ivanov et al., 2024; Zhou, Zhang, & Chan, 2024). Thus, we propose:

H3: Generative AI risk perception negatively moderates the relationship between generative AI trust and generative AI acceptance.

3. Method

3.1. Participants

This study focused on university students enrolled in 3 Chinese universities in 2025. For the sample size, we use G*Power 3.1.9.7 to calculate the minimum sample size, and the calculation result showed that the minimum sample size of this study was 138 (effect size $f^2 = 0.15$, α err prob. = 0.05, Power = 0.95). We conducted this survey in January 2025, employing convenience sampling as a non-probabilistic method reliant on high accessibility, voluntary participation, and the convenience of the participants (Lo et al., 2023). Stratton (2021) observed that researchers face inherent challenges in acquiring comprehensive data when studying specialized populations, leading them toward convenience sampling. We used the Wenjuanxing platform to create an online survey questionnaire and then shared the link of the online survey with participants via social media such as QQ and WeChat for them to fill out.

This online survey is conducted anonymously to better protect the privacy of the participants, and they can exit the survey at any time during the process. The study adheres to the ethical principles outlined in the Helsinki Declaration and the local ethics committee. The informed consent is obtained from all research participants. This online survey obtain 353 questionnaires among students from three universities in China. Samples with missing values and those with completion times of less than one minute were excluded. Ultimately, 292 valid questionnaires were obtained, resulting in an effective response rate of 82.72 %. The sample distribution indicates that there were 108 males (36.99 %) and 184 females (63.01 %). Among the respondents, 246 had rural household registration (84.25 %), while 46 had urban household registration (15.75 %). In terms of academic discipline distribution, there were 19 students from humanities and social sciences (6.51 %), 99 from science and engineering (33.90 %), 61 from agriculture (20.89 %), 17 from management (5.82 %), and 96 from medicine (32.88 %).

3.2. Measure

To assess university students' self-efficacy in generative AI, we adapted the AI Self-efficacy Scale developed by Wang and Chuang (2024) by replacing the term "AI" with "Generative AI" in the items. This scale comprises 22 items (e.g., "Some generative AI technologies/products make learning easier" and "I find that generative AI technologies/products are helpful for learning"), encompassing four dimensions: Assistance, Anthropomorphic Interaction, Comfort with AI, and Technological Skills. The responses were measured using a 7-point Likert scale, ranging from 1 (strongly disagree) to 7 (strongly agree). Higher scores indicate a greater self-efficacy in generative AI among college students. Participants with high generative AI self-efficacy feel confident and comfortable when interacting with generative AI that exhibits anthropomorphic features, and they regard the generative AI as a collaborative partner. In contrast, participants with low generative AI self-efficacy treat the generative AI merely as a tool, lack understanding and trust in it, and exhibit maladaptive behaviors during interaction. This scale shows high structural validity and internal consistency in previous study (Lintner, 2024). The scale demonstrated acceptable reliability and validity in this study, with $\alpha = 0.96$, $\chi^2/df = 4.16$, GFI = 0.78, and AGFI = 0.73.

To measure university students' trust in generative AI, we modified the ChatGPT Trust Scale created by Baek and Kim (2023) by substituting "ChatGPT" with "Generative AI". This scale consists of three items (e.g., "Generative AI is believable," "Generative AI is credible," and "Generative AI is trustworthy"). Responses were evaluated on a 7-point Likert scale, from 1 (strongly disagree) to 7 (strongly agree), with higher scores reflecting a greater level of trust in generative AI among college students. The scale exhibited strong reliability and validity in this study, with $\alpha = 0.95$, χ^2/df = N/A, GFI = 1.00, and AGFI = N/A.

To evaluate university students' perception of risks associated with generative AI, we adapted the AI Risk Perception Scale revised by Schwesig et al. (2023) by replacing "AI" with "Generative AI". This scale includes nine items (e.g., "When you think about this use of generative AI, how concerned, if at all, are you?" and "When you think of this use of generative AI, to what extent are you concerned?"). Responses were measured using a 5-point Likert scale, from 1 (not at all likely) to 5 (extremely likely), where higher scores indicate a stronger perception of risk regarding generative AI among college students. The scale demonstrated good reliability and validity in this study, with $\alpha = 0.95$, $\chi^2/df = 4.71$, GFI = 0.93, and AGFI = 0.86.

The Generative AI Acceptance Scale developed by Yilmaz et al. (2024) was employed to assess university students' acceptance of generative AI. This scale consists of 20 items (e.g., "I find generative AI applications useful in my daily life" and "The use of generative AI applications increases my chances of achieving the things that are important to me."). Responses were gathered using a 5-point Likert scale, from 1 (strongly disagree) to 5 (strongly agree), with higher scores indicating a greater acceptance of generative AI among college students. The scale exhibited good reliability and validity in this study, with $\alpha = 0.96$, $\chi^2/df = 3.53$, GFI = 0.84, and AGFI = 0.80.

3.3. Data analysis

Data entry and preprocessing of the questionnaire data were conducted using SPSS 27.0. Descriptive statistics were performed for all variables; categorical variables were summarized using frequencies and percentages, while continuous variables were reported as means and standard deviations. The independent samples *t*-test and one-way analysis of variance, respectively, examined the differences in the acceptance of generative AI among university students in terms of gender and major. Pearson correlation analysis was employed to explore the preliminary correlations among the research variables. Mediation effects were tested using the Bootstrap method in Amos 29.0; if the mediation effect value falls within the confidence interval and the interval does not include 0, it indicates the presence of a mediation effect. Moderation effects were examined using Model 1 of SPSS PROCESS 4.0, with a significance level set at $P < .05$.

4. Results

4.1. Gender and major differences in generative AI acceptance

The analysis results in Table 1 indicate that male students scored 3.63 on generative AI acceptance, while female students scored 3.59, with no significant gender-based difference observed ($t = 0.65$, $P > .05$). Among university students across different majors, generative AI acceptance scores were 3.87 for Humanities and Social Sciences, 3.57 for Science and Engineering, 3.66 for Agricultural Sciences, 3.59 for Management Studies, and 3.54 for Medical Sciences. No statistically significant differences in generative AI acceptance were found across these disciplinary categories ($F = 1.90$, $P > .05$).

4.2. Descriptive statistics and correlation analysis

Table 2 presents the correlation analysis results among the variables related to generative AI. The findings indicate a positive correlation between generative AI self-efficacy and generative AI trust ($r = 0.66$, $P < .01$). Conversely, there is a negative correlation between generative AI self-efficacy and generative AI risk perception ($r = -0.18$, $P < .01$). Additionally, generative AI self-efficacy is positively correlated with generative AI acceptance ($r = 0.59$, $P < .01$).

Further analysis reveals that assistance shows a positive correlation with generative AI trust ($r = 0.46$, $P < .01$), a negative correlation with generative AI risk perception ($r = -0.18$, $P < .01$), and a positive correlation with generative AI acceptance ($r = 0.57$, $P < .01$). The anthropomorphic interaction is positively correlated with both generative AI trust ($r = 0.59$, $P < .01$) and generative AI acceptance ($r = 0.41$, $P < .01$). Moreover, comfort with generative AI demonstrates a positive correlation with generative AI trust ($r = 0.58$, $P < .01$), a negative correlation with generative AI risk perception ($r = -0.22$, $P < .01$), and a positive correlation with generative AI acceptance ($r = 0.54$, $P < .01$). Finally, technological skills are positively correlated with generative AI trust ($r = 0.64$, $P < .01$), negatively correlated with generative AI risk perception ($r = -0.15$, $P < .01$), and positively correlated with generative AI acceptance ($r = 0.48$, $P < .01$).

4.3. Analysis of the mediating effect of generative AI trust and the moderating effect of generative AI risk perception

As shown in Fig. 1 and Fig. 2, generative AI self-efficacy is positively association with generative AI trust ($\beta = 0.66$, $P < .001$) and generative AI acceptance ($\beta = 0.49$, $P < .001$). Generative AI trust is positively association with generative AI acceptance ($\beta = 0.15$, $P < .05$). Anthropomorphic interaction is positively association with generative AI trust ($\beta = 0.25$, $P < .001$), and technological skills also are association with generative AI trust ($\beta = 0.39$, $P < .001$). Assistance is positively association with generative AI acceptance ($\beta = 0.34$, $P < .001$), while comfort with generative AI is positively association with generative AI acceptance ($\beta = 0.19$, $P < .05$). Additionally, technological skills are positively association with generative AI acceptance ($\beta = 0.16$, $P < .05$).

Table 3 shows that generative AI trust mediates the relationship between generative AI self-efficacy and generative AI acceptance, with an effect value of 0.10 and a confidence interval of [0.004, 0.19]. Additionally, generative AI trust mediates the relationship between anthropomorphic interaction and generative AI acceptance, with an effect value of 0.01 and a confidence interval of [0.01, 0.09]. Furthermore, generative AI trust also mediates the relationship between technological skills and generative AI acceptance, with an effect value of 0.01 and a confidence interval of [0.01, 0.14].

As shown in Table 4 and Fig. 3, generative AI trust is positively association with generative AI acceptance, while generative AI risk perception is positively association with generative AI acceptance. Moreover, the generative AI trust × generative AI risk perception negatively association with generative AI acceptance. Clearly, generative AI risk perception weakens the trust-acceptance link ($\beta = -0.08$, $P < .05$), suggesting high risk negates trust benefits.

Table 1
Analysis of the gender and major differences in generative AI acceptance.

	N	*M*	*SD*	*T/F*	*P*
Gender					
Male	108	3.63	0.61	0.65	0.52
Female	184	3.59	0.45		
Major					
Humanities and social sciences	19	3.87	0.43	1.90	0.11
Science and engineering	99	3.57	0.55		
Agricultural sciences	61	3.66	0.54		
Management studies	17	3.59	0.59		
Medical sciences	96	3.54	0.44		

5. Discussion

In this study, we designed and tested a model of generative AI acceptance, which includes generative AI self-efficacy, generative AI trust, generative AI risk perception, and generative AI acceptance. We found that university students with higher self-efficacy in generating AI have a higher acceptance of generative AI. The factors of assistance, comfort with generative AI, and technological skills within generative AI self-efficacy also are positively association with generative AI acceptance. Firstly, This may be attributed to the ability of generative AI to assist university students in learning professional knowledge, acquiring information resources, and writing papers (Gasaymeh et al., 2024; Golding et al., 2024). Secondly, when students interact smoothly and comfortably with generative AI products, they are more likely to accept and use generative AI (Park & Kim, 2023; Wang & Yu, 2024). Thirdly, having a good background in generative AI knowledge and confidence in using generative AI, university students reported a higher acceptance of generative AI (Bandura, 1982; Wang & Chuang, 2024).

Research by Baek and Kim (2023) found that trust in ChatGPT mediates the relationship between motivation to use ChatGPT and the intention to continue using ChatGPT; however, they did not find the association between the motivation to use ChatGPT and the intention to continue using it. Our study confirmed a positive association between motivational factors (generative AI self-efficacy) and the acceptance of generative AI, filling a gap in this area of study. It is worthy noticed that Tully et al. (2023) found that lower AI literacy predicts higher acceptance of artificial intelligence. This because of when ChatGPT was first introduced, most students exhibited limited AI literacy but were drawn to adopt and utilize the technology, primarily due to the novelty and mystique surrounding generative AI (Tully et al., 2023). However, as students progressively develop a deeper understanding of generative AI, its perceived mystique diminishes, reducing its initial attractiveness (Chen, Liu, & Liu, 2024; Tully et al., 2023). Only students who recognize generative AI's practical utility in enhancing academic task performance or providing emotional support are more likely to accept and use generative AI (Gasaymeh et al., 2024; Golding et al., 2024; Kim et al., 2025; Saklaki & Gardikiotis, 2024). This suggests that students with higher AI literacy are more inclined to adopt and utilize generative AI effectively over time (Chen, Tallant, & Selig, 2024; Laupichler et al., 2024; Wang et al., 2024). Therefore, the current show a positive association between generative AI self-efficacy and generative AI acceptance.

Although the use of generative AI can enhance self-efficacy.

Table 2
Results of descriptive statistics and correlation analysis.

Variables	*M*	*SD*	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*
1. Generative AI self-efficacy	5.20	0.88	1							
2. Assistance	5.66	0.85	0.80**	1						
3. Anthropomorphic interaction	4.78	1.29	0.85**	0.51**	1					
4. Comfort with generative AI	5.19	0.99	0.91**	0.66**	0.71**	1				
5. Technological skills	4.95	1.14	0.82**	0.53**	0.63**	0.71**	1			
6. Generative AI trust	4.69	1.06	0.66**	0.46**	0.59**	0.58**	0.64**	1		
7. Generative AI risk perception	2.83	0.70	−0.18**	−0.18**	−0.08	−0.22**	−0.15*	−0.13*	1	
8. Generative AI acceptance	3.60	0.51	0.59**	0.57**	0.41**	0.54**	0.48**	0.47**	−0.21**	1

** Note: $P < .01$.

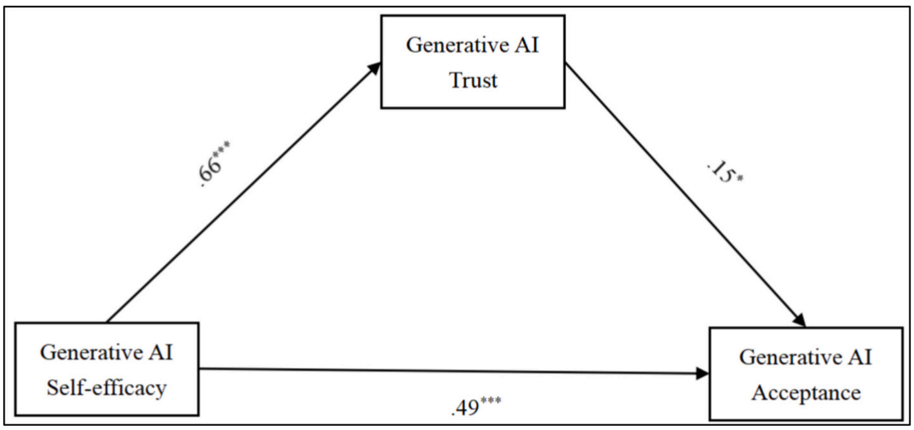

Fig. 1. The graph of the path analysis results (1).

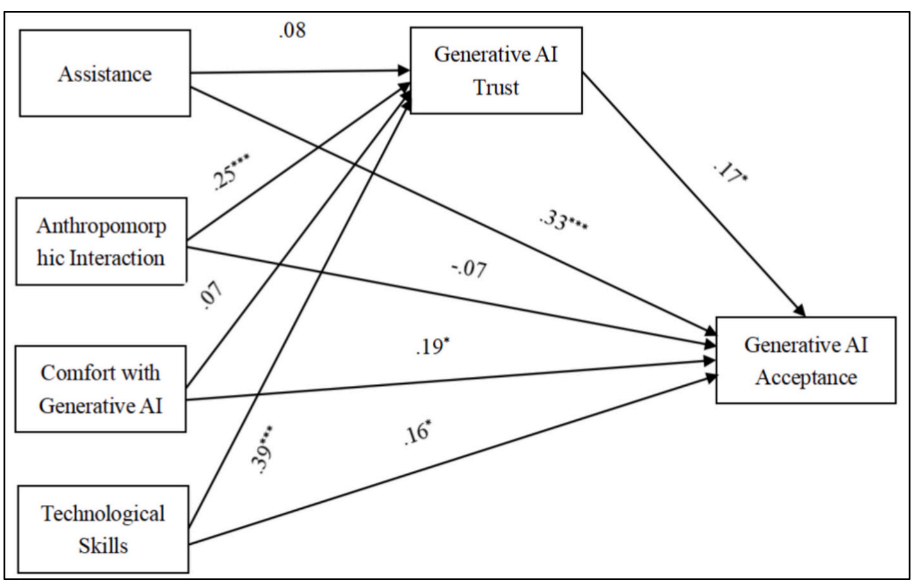

Fig. 2. The graph of the path analysis results (2).

However, the use of generative AI has varying effects on users' creativity. For instance, Doshi and Hauser (2024) found that when it comes to story creation, generative AI helps weaker authors improve their creativity. Generative AI is especially beneficial for less-skilled authors, a phenomenon that has also been observed in studies from other fields. Noy and Zhang (2023) found that generative AI can benefit workers with lower efficiency.

Our study also found that generative AI trust mediates the relationship between generative AI self-efficacy and generative AI acceptance. Previous research has primarily examined the relationship between trust in AI and the intention to use AI (Baek & Kim, 2023); however, this study further explores the relationships among generative AI self-efficacy, generative AI trust, and generative AI acceptance. Specifically, anthropomorphic interaction and technological skills indirectly predict university students' acceptance of generative AI through trust in generative AI. This may be due to the fact that, during their use of generative AI, anthropomorphic interactions make a positive user experience (Davis, 1989). Consequently, they may develop greater trust in generative AI and utilize it to assist them in completing their learning tasks (Baek & Kim, 2023; Kanont et al., 2024; Song et al., 2022). Moreover, when

Table 3
Results of mediating effect.

Path	*Mediating values*	*95 % CI*
Generative AI self-efficacy → Generative AI trust → Generative AI acceptance	0.10	[0.004, 0.19]
Anthropomorphic interaction → Generative AI trust → Generative AI acceptance	0.01	[0.01, 0.09]
Technological skills → Generative AI trust → Generative AI acceptance	0.01	[0.01, 0.14]

Table 4
Results of moderating effect.

Dependent variable: Generative AI acceptance	*Coeff*	*se*	*t*	*P*	*95 % CI*
Constant	1.66	0.51	3.26	< 0.001	[0.66, 2.67]
Generative AI trust	0.44	0.09	4.83	< 0.001	[0.26, 0.62]
Generative AI risk perception	0.32	0.18	1.83	0.07	[−0.02, 0.67]
Generative AI Trust × Generative AI risk perception	−0.08	0.03	−2.51	< 0.05	[−0.14. -0.02]
R-sq = 0.26					
F = 34.00					

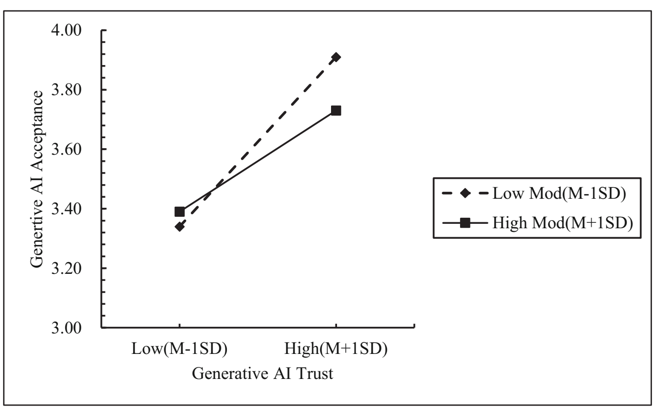

Fig. 3. The resulting graph of the moderating effect of generative AI risk perception.

university students possess both the background knowledge supporting the use of generative AI and a robust level of confidence in its application, they will trust generative AI and employ it to accomplish tasks (Baek & Kim, 2023; Bandura, 1982).

It is noteworthy that perceived risk in generative AI negatively moderates the relationship between trust in generative AI and acceptance of generative AI. This may be attributed to university students' awareness of potential risks and serious consequences associated with using generative AI, which can diminish their acceptance of it (Zhu et al., 2024). For example, if the information obtained through generative AI is inaccurate or if errors occur in the computer code generated, it could lead to mistakes in students' learning tasks (Johnston et al., 2024; Zhu et al., 2024). Additionally, generative models require vast amounts of copyrighted texts, images, and audio to be used for data training, which may lead to copyright disputes and the potential for academic misconduct (Ivanov et al., 2024; Zhu et al., 2024). Based on such concerns, students lower their trust in and acceptance of generative AI.

5.1. Implications

Based on the Social Cognitive Theory (SCT), Technology Acceptance Model (TAM), and previous finds, this study constructs a model for generative AI acceptance among university students. Regarding contributions, firstly, this study enriches the research on artificial intelligence based on the social cognitive theory and technology acceptance model. Secondly, our research identifies generative AI self-efficacy as a significant predictor of generative AI acceptance and confirmed the mediating of generative AI trust and moderating of generative AI risk perception. Thirdly, previous research based on the Technology Acceptance Model (TAM) has demonstrated a correlation between technological self-efficacy and acceptance. For example, Xu et al. (2024) showed that perceived technological self-efficacy is associated with the use of information and communication technologies. In contrast to these traditional TAM applications, the present study investigates the relationship between the four specific dimensions of generative AI self-efficacy (assistance, anthropomorphic interaction, comfort with generative AI, and technological skills) and the acceptance of generative AI. This provides new insights into how the Technology Acceptance Model can be used to understand the relationship between generative AI self-efficacy and acceptance.

5.2. Limitations and directions for future research

Most of our research hypotheses were supported; however, there are still some limitations in our study. First, this study focuses exclusively on Chinese university students, primarily those majoring in STEM-related fields. Because STEM students tend to be more familiar with technology, this may limit the generalizability of the findings. Future research could employ a more systematic sampling strategy to include students from various disciplines, compare different majors, or even contrast student populations with non-student groups to uncover additional insights. Second, the geographical coverage of our survey sample was not extensive. Future studies could broaden the survey area to understand the acceptance of generative AI products and their influencing factors among users from different geographical regions. Third, this study utilized self-report scales to measure generative AI self-efficacy, generative AI trust, generative AI risk perception, and generative AI acceptance among university students. This self-report method carries inherent subjectivity, potentially leading to less accurate measurements of the research variables. Future studies could adopt experimental methods to achieve more accurate measurements of these variables and better uncover the relationships between generative AI self-efficacy, trust, risk perception, and acceptance. Fourth, the reliability and validity of the AI-related scales used in this study still have some flaws; future research could further validate and refine these scales. This study applied the Generative AI Self-efficacy Scale to Chinese university students, but the validity showed some shortcomings. This may be because our sample was mainly composed of students from western China, without including students from eastern China, which could affect the scale's validity. Since there are clear economic development differences between western and eastern regions of China, the development and adoption of artificial intelligence also vary, and consequently the usage patterns of generative AI among university students are likely to differ. Future research should recruit participants from both western and eastern universities to better validate the Generative AI scale within the broader Chinese student population. In addition, the Generative AI Trust Scale consisting of three items, although having good reliability, may involve more dimensions of trust in Generative AI. Measuring only with three items is not comprehensive enough. Future research can develop a more comprehensive Generative AI Trust Scale for measurement to obtain more accurate results.

Fifth, the current research design omitted the participants' pre-existing generative AI experience, which may have an impact on the acceptance of generative AI. Subsequent studies could further control participants' pre-existing generative AI experience to obtain more accurate relationships between the study variables. Sixth, the correlation between generative AI self-efficacy and AI acceptance may be moderated by the task complexity. That is to say, when tackling tasks with high complexity, generative AI becomes less effective in assisting people. This consequently weakens their generative AI self-efficacy and reduces their acceptance of generative AI. Future research can use experimental designs to investigate how task difficulty moderates the relationship between generative AI self-efficacy and acceptance of generative AI. Finally, although this study measured generative AI in principle, it did not conduct in-depth measurement of different types of generative AI. This may lead to situations where participants, when reporting their attitudes toward generative AI via scales, might have different types of generative AI in mind, resulting in potential bias in the reported results. Future research can investigate and compare differences across various types of generative AI to gain richer insights.

CRediT authorship contribution statement

Gen Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tingrong Yu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical statement

This study obtained informed consent from all research participants. The study adheres to the ethical principles outlined in the Helsinki Declaration and the local ethics committee.

Funding

No funding was received for this study.

Declaration of competing interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this research article.

Data availability

The data that support the findings of this study are available on request from the corresponding author.

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